

PAPER_701: UQFF Knowledge Base (KB1-KB19): Red Dwarf Compression Paper Assimilation

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Session: 174 | **Version:** v5.31

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Abstract

The UQFF Knowledge Base (KB1-KB19) represents 19 foundational theoretical papers assimilated through the Red Dwarf Compression framework. Core topics include the inertial operator, caduceus coil twist, Solfeggio harmonic resonances, pseudo-monopole fields, and dark energy power quantization.

1. KB1: Quantum Wavefunction and Inertial Operator

$$\Psi(r, \theta, \phi, t) = A Y_{lm}(\theta, \phi) \frac{\sin(kr - \omega t)}{r} e^{-\alpha|r-r_0|}$$

Caduceus coil twist: $\phi_{twist} = \beta \sin(\omega t)$, $\beta \approx 0.9999$

Inertial operator: $\hat{O}\Psi = \lambda_I \left(\frac{\partial}{\partial t} + i\omega_m \mathbf{r} \times \nabla \right) \Psi$

2. KB2: Solfeggio Harmonic Resonance

Frequencies $f \in \{174, 285, 396, 417, 528, 639, 741, 852, 963\}$ Hz

$$E_{solfeggio} = \sum_n \hbar \omega_n = \sum_n h f_n$$

3. KB3: Pseudo-Magnetic Monopole

$$B_{pseudo}(r) = \frac{\mu_0 q_m}{4\pi r^2}$$

4. KB4: Dark Energy Power

$$P_{DE} = \rho_{SCm} c^2 V_{cosmos} / t_H = 7.09 \times 10^{-51} \text{ W}$$

5. KB: Composite UQFF Term

$$g_{KB} = \hat{O}\Psi(r, t) + E_{solfeggio} \frac{\rho_{SCm}}{\rho_{UA}} + B_{pseudo} k + P_{DE} \frac{1 + f_{TRZ}}{k_B T_{CMB}}$$

6. Constants (UQFF v5.31)

Constant	Value
ρ_{UA}	$7.09 \times 10^{-36} \text{ J/m}^3$
ρ_{SCm}	$7.09 \times 10^{-37} \text{ J/m}^3$
f_{TRZ}	0.1
κ	$5 \times 10^{-4} \text{ day}^{-1}$
$[SSq]$	0.57
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$

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$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Red Dwarf Compression Papers KB1–KB19
- UQFF Framework, Star-Magic Session 174

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